



University of Engineering and Technology, Lahore

Department of Computer Science

Project Proposal

Team Members

Name	Roll Number	Email
Muhammad Uzair	2026(S)-CS-96	2026scs96@uet.edu.pk

Session: Spring'26 Afternoon

Section: B

Semester: 1st

Course: CSC101 - Discrete Mathematics

Teacher: Mr. Waqas Ali

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CSC101 - Discrete Mathematics

2026(S)-CS-96

Project Proposal

Submission Deadline: Monday, 31-5-2026, 11:59 PM

1. Project Title

Lottery Probability Calculator: Understanding Rarity Through Combinatorics

2. Selected Track

Track 6: Counting and Combinatorics (Rosen Chapter 6)

3. Chosen Project

Lottery Probability Calculator. The program accepts lottery parameters from the user, specifically the total number of balls and how many are drawn per ticket, and then computes the exact odds of winning the jackpot, the chances of matching only some of the numbers, the number of tickets theoretically needed to guarantee a win, and the expected monetary value of playing. Every result is shown step by step using the combination formula so the user can follow the reasoning rather than just reading a final number.

4. Problem Statement

People talk about the lottery constantly but almost nobody has a real feel for how astronomically unlikely winning it actually is. The same gap in intuition shows up across naturally rare events in biology, physics, and everyday life. We use phrases like "one in a million" without truly grasping what that scale means in practice.

The core challenge this project addresses is making those odds tangible. When a user enters six numbers out of forty-nine and the program responds with a probability of roughly 1 in 13,983,816, and then shows that buying one ticket every week would take on average around 268,920 years before a win, the number suddenly means something real. The project turns abstract probability into a concrete, almost visceral understanding of rarity.

5. Motivation

There is something genuinely captivating about how rare most things in life are when you actually sit down and count them out. The fact that a shuffled deck of cards is almost certainly in an arrangement that no human being has ever seen before, that the specific combination of biological events that produced any living creature borders on impossible, that something as mundane as matching six numbers on a lottery ticket connects directly to one of the deepest ideas in mathematics. These are not just dry statistics. They are perspective-shifting realizations.

The lottery is one of the most universally accessible entry points into this world of combinatorics because nearly everyone is familiar with it, and most people hold a vague sense that winning is very unlikely without ever confronting the actual figure. Building something that lays this out clearly, line by line, felt like the most honest way to make Chapter 6 material feel real rather than abstract.

6. Mathematical Foundations

The project centers on the combination formula $C(n, r) = n! \text{ divided by } (r! \text{ multiplied by } (n \text{ minus } r)!)$, drawn from Chapter 6 of Rosen's Discrete Mathematics and Its Applications. This formula counts the number of ways to choose r items from n when order does not matter, which is exactly the structure of a lottery draw.

Jackpot probability is then simply 1 divided by $C(n, r)$, since only one combination wins out of all possible ones. For partial match probabilities, the formula becomes $C(k, m)$ multiplied by $C(n \text{ minus } k, r \text{ minus } m)$, divided by $C(n, r)$, where k is the number of winning balls drawn, m is how many the ticket matched, and the second factor counts the ways to fill the remaining spots with non-winning numbers. Expected value is computed as the prize amount multiplied by the win probability, minus the cost of one ticket.

7. Planned Approach

The implementation will be in C++, chosen for its efficiency with integer arithmetic and the absence of any dependency on external libraries for the core calculations.

The combination formula will be computed iteratively rather than by calculating full factorials. The trick is to simplify the fraction progressively during multiplication, which keeps intermediate values from growing to sizes that overflow a 64-bit integer. For edge cases involving very large numbers, the program uses division at each step to prevent overflow before it can occur.

The user will be presented with a numbered menu where they input the lottery configuration and then choose which calculation to run. Each result screen prints the formula first, substitutes the values next, and then shows the final answer, so the user sees every step. A comparison mode allows two different lottery types to be entered side by side so their odds can be contrasted directly.

Testing will be done by cross-checking all outputs against officially published lottery probability figures, for instance confirming that the 6-from-59 configuration of the UK National Lottery produces the stated odds of 1 in 45,057,474.

8. Expected Outcomes

The finished program will accept any lottery configuration and output the jackpot probability, partial match probabilities for every possible matching level, the theoretical number of tickets needed to guarantee a win, the expected monetary value of a single ticket for a given prize, and a formatted side-by-side comparison between two lottery formats. All calculations will include the step-by-step working so a first-semester student can follow every line.

9. Potential Challenges

The main technical hurdle is integer overflow. The factorial of 49 is a 63-digit number, far beyond what a standard 64-bit integer can hold. The solution is to avoid computing raw factorials entirely and instead cancel common factors from numerator and denominator before multiplying, keeping intermediate values well within a safe range.

There is also a conceptual stretch involved. Some ideas used here, particularly expected value and the multi-match probability formula, go somewhat beyond what has been formally introduced in the course at this point in the semester. The plan is to work through the relevant parts of Rosen independently and annotate the corresponding code sections thoroughly so the logic is self-explanatory.

Finally, presenting the output in a way that is actually informative rather than just a column of numbers will require care. The goal is to print the symbolic formula first, then substitute real values, then state the answer in plain language, so every calculation tells a readable story rather than just producing a result.

10. Resources Required

No external libraries or datasets are needed beyond a standard C++17 compiler. The primary mathematical reference is Rosen's Discrete Mathematics and Its Applications, Chapter 6. Published official probability tables from major national lotteries will be used as verification benchmarks.

11. References

1. Rosen, K. H. Discrete Mathematics and Its Applications. Chapter 6: Counting.
2. UK National Lottery official probability documentation.
3. Powerball official probability tables.
4. cppreference.com, C++ standard library reference.

Declaration: I confirm that this proposal represents my own work and that I understand the academic integrity requirements outlined in the project guidelines.

Muhammad Uzair: _____

Date: 31 May
2026